DEEP LEARNING THE NANOSCALE: A PHYSICS-INFORMED 3D UNET FOR SOLVING THE THERMOPLASMONIC INVERSE PROBLEM

A PROJECT REPORT

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Kushagra Sinha

ABSTRACT

Metallic nanoparticles under laser illumination act as nanoscale heat sources through localized surface plasmon resonance. Quantitative phase imaging (QPI) techniques can map the resulting thermal landscape by measuring the optical phase shift accumulated through the heated medium. However, this measurement inherently projects the three-dimensional temperature field onto a two-dimensional map—all depth information is lost. Recovering the full volumetric distribution $T(x, y, z)$ from a single 2D phase image is therefore a severely ill-posed inverse problem.

In this work, we develop a complete framework to solve this inverse problem. We first construct an analytical forward model based on the thermal Green's function for a three-layer stratified medium (substrate–liquid–superstrate), using the method of image charges. This model generates paired datasets of 2D phase maps and 3D temperature tensors. We then design a physics-informed 3D U-Net that maps a 2D phase input to a full 3D thermal volume, regularized by enforcing the steady-state Laplace equation $\nabla^2 T = 0$ in source-free regions. We also develop an automated experimental data-cleaning pipeline for off-axis digital holographic interferometry. Preliminary results demonstrate accurate volumetric reconstruction with correct depth localization from single 2D phase inputs.

LIST OF SYMBOLS AND ABBREVIATIONS

Symbol	Description
σ_{abs}	Absorption Cross-Section
Q	Absorbed Heat Power (W)
$\Delta\phi$	Optical Phase Difference
κ	Thermal Conductivity
$G(\vec{r}, \vec{r}_0)$	Thermal Green's Function
β_i	Thermo-Optic Coefficient (order i)
Δn	Refractive Index Change
ΔT	Temperature Rise Above Ambient
R_{ij}, T_{ij}	Thermal Reflection / Transmission Coefficients
W	Convergence Parameter ($R_{\text{top}} \cdot R_{\text{bot}}$)
I	Laser Irradiance (W/m^2)

Abbreviation	Full Form
LSPR	Localized Surface Plasmon Resonance
TQSI	Thermal Quantitative Phase Shearing Interferometry
QPI	Quantitative Phase Imaging
PINN	Physics-Informed Neural Network
OPD	Optical Path Difference

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1. INTRODUCTION

1.1 Thermoplasmonics: Light to Heat at the Nanoscale

When a metallic nanoparticle is illuminated at its plasmon resonance, the free conduction electrons undergo collective oscillation against the ionic lattice. This localized surface plasmon resonance (LSPR) greatly enhances the particle's ability to absorb light Maier [2007]. The absorbed electromagnetic energy is rapidly converted to heat through electron–phonon coupling on a picosecond timescale Baffou, Berto, Ureña, Quidant, Monneret, Polleux, and Rigneault [2013], and the particle becomes a point-like heat source radiating thermal energy into its surroundings.

The heat power generated by a nanoparticle under continuous-wave illumination is

$$Q = \sigma_{\text{abs}}\, I, \tag{1.1}$$

where σ_{abs} is the absorption cross-section and I is the laser irradiance. By tuning the particle's size, shape, and material, one can engineer σ_{abs} to peak at a desired wavelength—making nanoparticles exquisite, sub-wavelength heat sources with applications spanning photothermal therapy, nanochemistry, and heat-assisted magnetic recording Govorov and Richardson [2007], Brongersma, Halas, and Nordlander [2015], Baffou [2017].

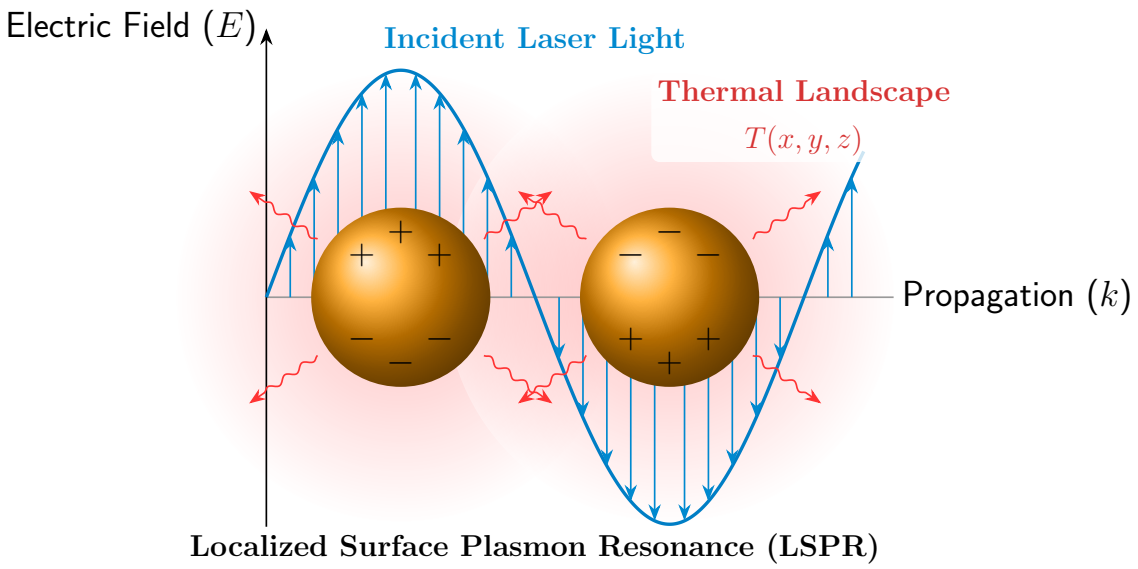

Fig. 1.1: The thermoplasmonic heating process. An incident laser excites an LSPR in metallic nanoparticles. Non-radiative plasmon decay leads to Joule heating, which diffuses outward creating a 3D thermal landscape $T(x, y, z)$.

1.2 Measuring Heat Optically: Quantitative Phase Imaging

Conventional thermometry (thermocouples, IR cameras) cannot resolve temperature distributions around individual nanoparticles. Instead, we exploit the *thermo-optic effect*: heating the medium changes its refractive index. For water, $dn/dT \approx -10^{-4}$ K^{-1}. A weak probe beam traversing the heated medium accumulates a phase shift proportional to the integrated refractive-index change along the optical axis Baffou, Bon, Savatier, Polleux, Zhu, Merlin, Rigneault, and Monneret [2012], Bon, Maucort, Wattellier, and Monneret [2009]:

$$\Delta\phi(x, y) = \frac{2\pi}{\lambda_{\text{probe}}} \int_{-\infty}^{\infty} \frac{dn}{dT}\, \Delta T(x, y, z)\, dz. \tag{1.2}$$

Quantitative phase imaging (QPI) techniques—such as quadriwave lateral shearing interferometry—extract this 2D phase map $\Delta\phi(x, y)$ with high sensitivity. However, Eq. 1.2 reveals a fundamental limitation: the probe beam integrates along z, collapsing all depth information. We obtain a 2D projection of a 3D reality.

1.3 The Inverse Problem

The *forward problem* is straightforward: given heat sources $\{Q_i\}$ at known positions, solve the heat equation for $\Delta T(x, y, z)$, then integrate along z to predict $\Delta\phi(x, y)$. The *inverse problem* asks the opposite: given a measured $\Delta\phi(x, y)$, recover the full 3D temperature

field.

This inverse problem is *ill-posed* in the sense of Hadamard Hadamard [1902]—infinitely many 3D temperature distributions can produce the same 2D phase map. Classical approaches based on Tikhonov deconvolution in Fourier space Tikhonov, Goncharsky, Stepanov, and Yagola [1995], Baffou, Ureña, Berto, Monneret, Quidant, and Rigneault [2014] can recover the 2D heat-source density, but they cannot resolve depth. Furthermore, they assume an infinite homogeneous medium, whereas real experiments involve layered geometries (glass/water/air) where the thermal conductivity changes discontinuously across interfaces.

Our approach replaces the classical deconvolution with a physics-informed neural network: a 3D U-Net trained on synthetic data from a rigorous forward model and regularized by the steady-state heat equation. The network learns to invert the projection and recover depth-resolved thermal information from a single 2D measurement. The physics enforced during training—$\nabla^2 T = 0$ outside the sources—constrains the solution space to thermodynamically valid fields. A detailed description of the computational pipeline is provided in the Methodology (Appendix A).

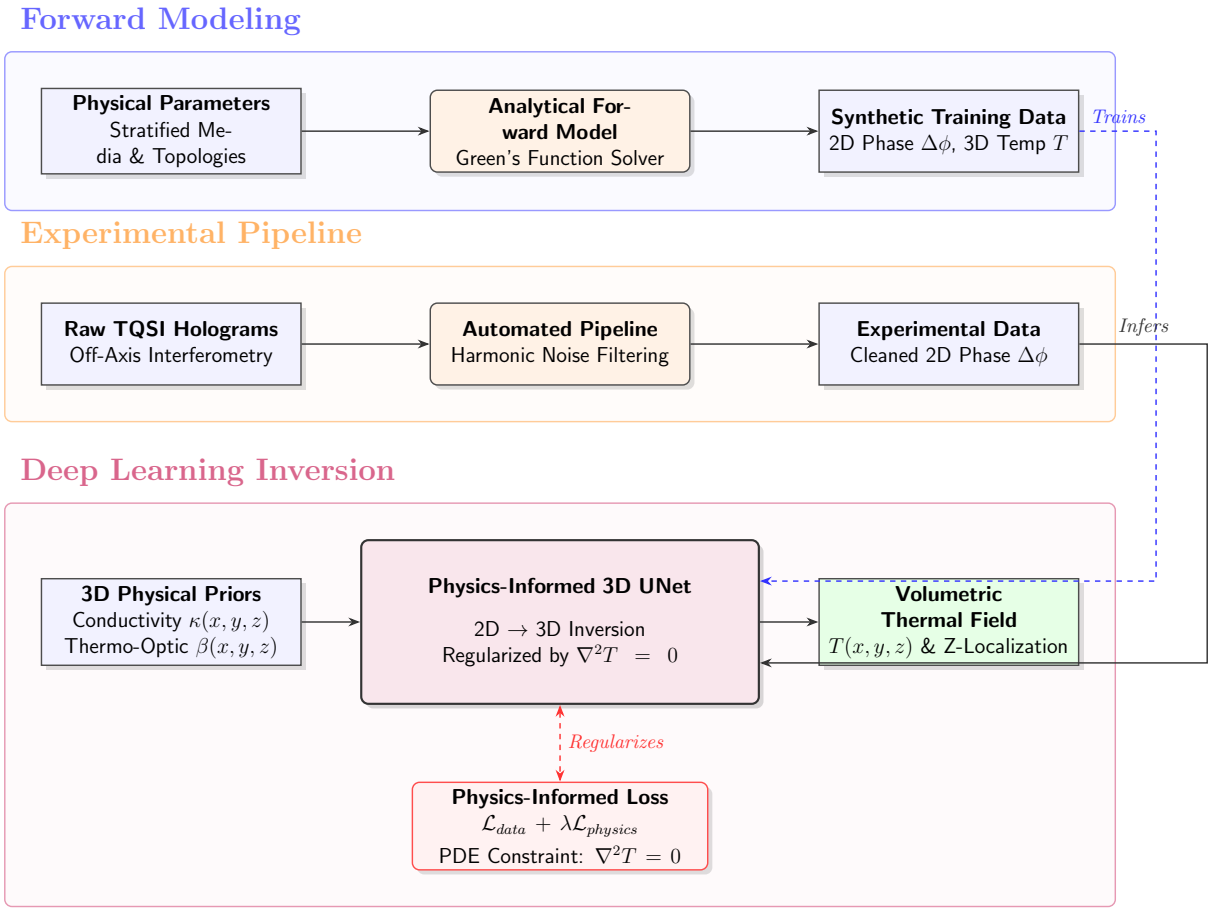

Fig. 1.2: Overview of the project workflow. The analytical forward model generates synthetic training data. Experimental phase maps are cleaned automatically. The physics-informed 3D UNet maps 2D phase to 3D temperature, constrained by $\nabla^2 T = 0$.

2. PHYSICS OF THERMOPLASMONIC HEATING

2.1 Optical Response of Metallic Nanoparticles

2.1.1 Polarizability and the Plasmon Resonance

Consider a metallic nanosphere of radius $R \ll \lambda$ in a uniform dielectric host (ε_m). The electric field is effectively constant across the particle (quasistatic limit), and Laplace's equation $\nabla^2\Phi = 0$ can be solved exactly for a sphere of complex dielectric function $\varepsilon(\omega)$ Maier [2007], Jackson [1999]. The result is a dipolar response with polarizability

$$\alpha = 4\pi R^3 \frac{\varepsilon - \varepsilon_m}{\varepsilon + 2\varepsilon_m}. \tag{2.1}$$

The denominator vanishes when $\mathrm{Re}(\varepsilon) = -2\varepsilon_m$ (Fröhlich condition), producing the LSPR. At resonance, the internal field is dramatically enhanced and the absorption cross-section peaks.

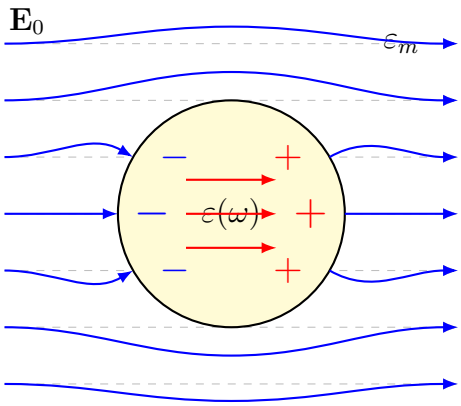

Fig. 2.1: Localized surface plasmon in the quasistatic picture. The applied field displaces the electron cloud, inducing a dipole that creates a uniform internal field and distorts the external field.

2.1.2 Optical Cross-Sections

The oscillating dipole radiates and absorbs light. The cross-sections follow from the polarizability Bohren and Huffman [1998]:

$$\sigma_{\text{ext}} = k\ \text{Im}(\alpha), \qquad \sigma_{\text{sca}} = \frac{k^4}{6\pi}|\alpha|^2, \qquad \sigma_{\text{abs}} = \sigma_{\text{ext}} - \sigma_{\text{sca}}, \tag{2.2}$$

where $k = 2\pi\sqrt{\varepsilon_m}/\lambda$. Since $\sigma_{\text{abs}} \propto R^3$ and $\sigma_{\text{sca}} \propto R^6$, small nanoparticles ($R \lesssim 20$ nm) are predominantly absorbers—ideal nano-heaters Maier [2007].

For particles beyond the quasistatic limit ($R \gtrsim 20$ nm), the full Mie theory provides exact cross-sections from expansions in vector spherical harmonics Mie [1908], Bohren and Huffman [1998]. For nanorods, the Mie–Gans extension with Kuwata's dynamic depolarization correction Kuwata, Tamaru, Esumi, and Miyano [2003] captures the longitudinal plasmon resonance tunability into the near-infrared.

2.2 Steady-State Heat Diffusion

2.2.1 The Point-Source Solution

Under continuous illumination, a nanoparticle reaches thermal steady state within nanoseconds Baffou et al. [2013]. The temperature rise $\Delta T = T - T_\infty$ obeys the Poisson equation $\kappa\nabla^2 T = -q(\mathbf{r})$, where q is the volumetric heat source. For a point source of power Q in a homogeneous medium of conductivity κ:

$$\Delta T(r) = \frac{Q}{4\pi\kappa\, r}, \tag{2.3}$$

with the maximum $\Delta T_{\text{max}} = Q/(4\pi\kappa R)$ at the particle surface. For N particles, the total field is a simple superposition: $\Delta T(\mathbf{r}) = \sum_i Q_i/(4\pi\kappa|\mathbf{r} - \mathbf{r}_i|)$.

2.2.2 Green's Function for a Three-Layer Medium

In realistic experiments, nanoparticles sit on a glass substrate (κ_1) immersed in a fluid (κ_2), possibly capped by a superstrate (κ_3). The conductivity mismatch at the interfaces redirects heat flow. We solve this using the *method of image charges*—analogous to electrostatic image problems Baffou et al. [2014], Baffou, Quidant, and Girard [2010].

At each interface between media i and j, we define thermal reflection and transmission coefficients:

$$R_{ij} = \frac{\kappa_i - \kappa_j}{\kappa_i + \kappa_j}, \qquad T_{ij} = \frac{2\kappa_i}{\kappa_i + \kappa_j}. \tag{2.4}$$

For a point source at (x_0, y_0, z_0) in the middle layer (bounded by b_1 and b_2, with $d = b_2 - b_1$), the "image charges" bounce between the two interfaces. The loop-gain factor

$W = R_{\text{top}} R_{\text{bot}}$ satisfies $|W| < 1$ for all real materials, guaranteeing convergence. Defining the inverse-distance kernel $G_0(z, \zeta) = (\rho^2 + (z-\zeta)^2)^{-1/2}$ with $\rho = \sqrt{(x - x_0)^2 + (y - y_0)^2}$, the temperature in the middle layer is:

$$T_2 = \frac{Q}{4\pi\kappa_2}\left[G_0(z, z_0) + \sum_{n=0}^{\infty} W^n\Big(W G_0(z, \zeta_1^{(n)}) + R_{\text{top}}\, G_0(z, \zeta_2^{(n)}) + R_{\text{bot}}\, G_0(z, \zeta_3^{(n)}) + W G_0(z, \zeta_4^{(n)})\Big)\right], \tag{2.5}$$

with image depths $\zeta_1^{(n)} = z_0 + 2(n+1)d$, $\zeta_2^{(n)} = 2b_2 - z_0 + 2nd$, $\zeta_3^{(n)} = 2b_1 - z_0 - 2nd$, $\zeta_4^{(n)} = z_0 - 2(n+1)d$. The fields transmitted into the substrate ($z < b_1$) and superstrate ($z > b_2$) take analogous forms with the appropriate transmission coefficients T_{21} and T_{23}.

2.3 From Temperature to Optical Phase

The refractive index of water depends nonlinearly on temperature Baffou, Rigneault, Marber, and Quidant [2020]:

$$\Delta n(\Delta T) = \beta_1\, \Delta T + \beta_2\, \Delta T^2 + \beta_3\, \Delta T^3 + \beta_4\, \Delta T^4, \tag{2.6}$$

where the β_k are the thermo-optic coefficients centered at the ambient temperature T_∞. The measured optical phase shift is

$$\Delta\phi(x, y) = \frac{2\pi}{\lambda_{\text{probe}}} \int_0^h \Delta n(x, y, z)\, dz, \tag{2.7}$$

where h is the sample thickness. This equation constitutes the *forward model*: given $\Delta T(x, y, z)$, compute $\Delta\phi(x, y)$.

3. FORWARD MODEL AND SIMULATIONS

3.1 Green's-Function Forward Solver

The synthetic cases use the supplied three-layer Green's-function solver. For an isolated absorber in a homogeneous medium, the far-field steady-state temperature rise has the standard thermoplasmonic form

$$\Delta T_i(r) = \frac{\sigma_{\mathrm{abs},i} I_i}{4\pi\kappa r}, \qquad \Delta T(\mathbf{r}) = \sum_i \Delta T_i(|\mathbf{r} - \mathbf{r}_i|), \tag{3.1}$$

with the finite particle radius used to regularize the near field. This is the Green's-function expression used as a baseline in thermoplasmonic modelling Baffou et al. [2010, 2013]. The implemented solver adds the two planar interfaces through an image-source construction and evaluates the temperature over a $128 \times 128 \times 32$ volume.

The linear optical phase kernel is similarly evaluated in Fourier space:

$$G_\phi(R) = \frac{\beta_1}{4\pi\kappa_{\mathrm{eff}}} \operatorname{arcsinh}\left(\frac{h}{R}\right). \tag{3.2}$$

Higher-order thermo-optic corrections are computed via direct z-integration of the nonlinear residual $\Delta n_{\mathrm{res}}(T_z) = \beta_2 T_z^2 + \beta_3 T_z^3 + \beta_4 T_z^4$.

3.2 Representative Simulations

The forward solver generates high-fidelity synthetic datasets spanning diverse nanoparticle topologies, materials, and illumination profiles. Figures 3.1–3.3 illustrate representative examples.

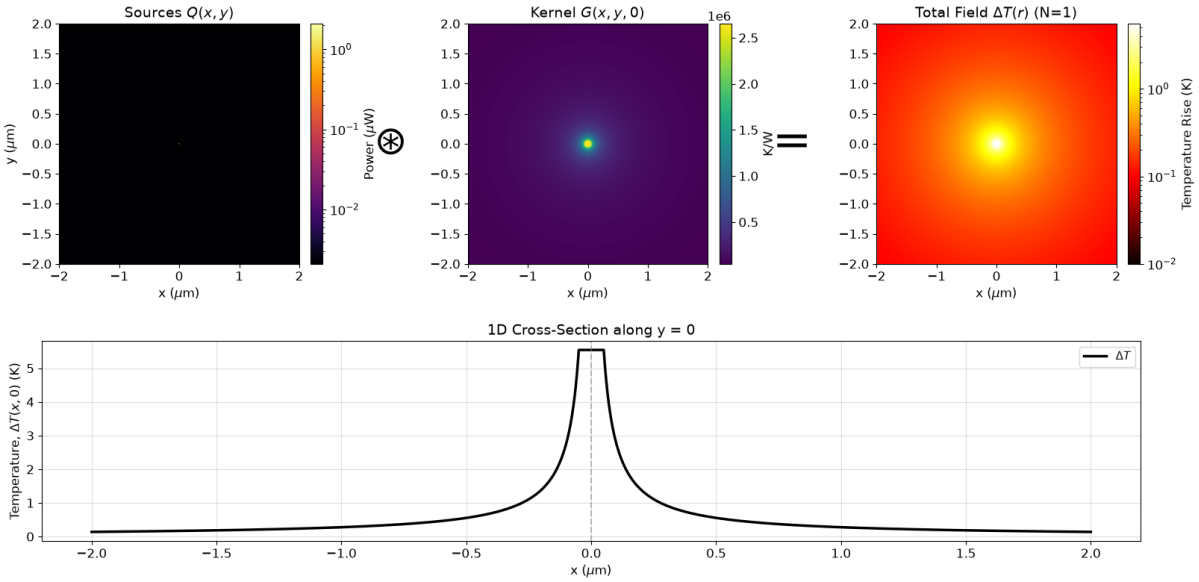

Fig. 3.1: Single isolated nanoparticle. Au nanosphere ($a = 50$ nm) in water ($\kappa = 0.60$ W/mK), $\lambda = 532$ nm, $I_0 = 100$ mW/μm^2. Mie-theory $\sigma_{\text{abs}} = 2.09 \times 10^{-2}$ μm^2. The thermal kernel $G(x, y, 0)$ is bounded at $r = a$ to prevent a central singularity. The cross-section shows the $1/r$ thermal decay.

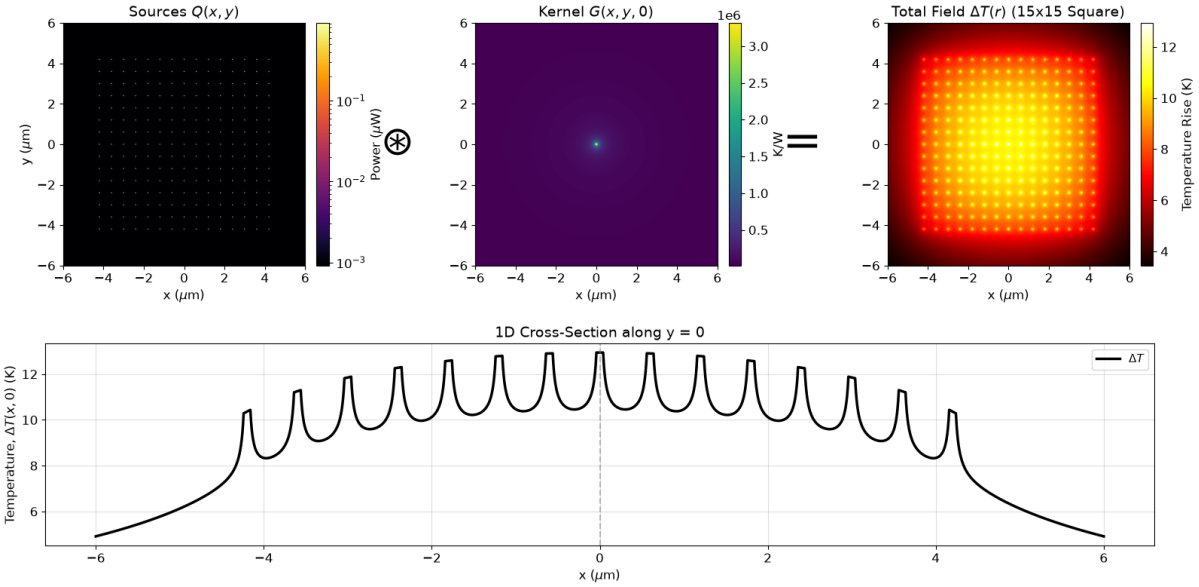

Fig. 3.2: Square nanoparticle array (15×15 Au spheres, $a = 40$ nm, pitch 600 nm) under uniform illumination at 50 mW/μm^2. The collective heating envelope is superimposed with localized intra-particle thermal spikes reaching $\Delta T \approx 13$ K.

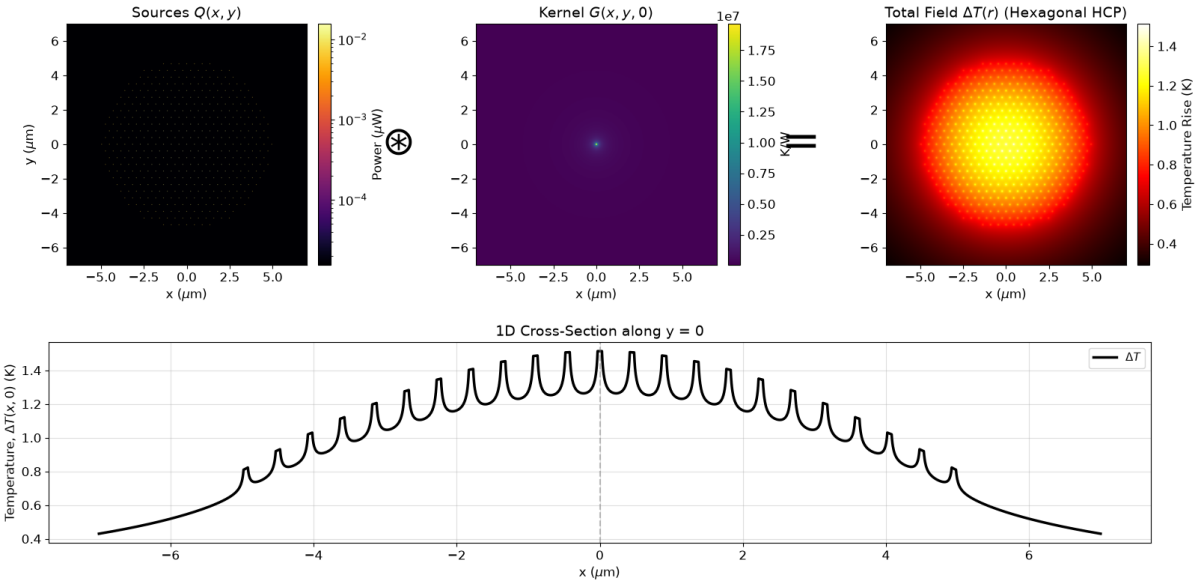

Fig. 3.3: Hexagonal lattice of Ag spheres ($a = 30$ nm, pitch 450 nm) in isopropanol ($\kappa = 0.135$ W/mK) under Gaussian illumination ($w_0 = 8$ μm). The low thermal conductivity of IPA strongly amplifies the collective thermal envelope.

3.3 Forward Model Configurations

The analytical solver was evaluated on four three-layer configurations that isolate the influence of the two thermal interfaces:

1. **Homogeneous** ($\kappa_1 = \kappa_2 = \kappa_3$): Perfect $1/r$ decay.
2. **Substrate-supported** ($\kappa_1 \neq \kappa_2 = \kappa_3$): Asymmetric z-profile; heat preferentially diffuses into high-κ glass.
3. **Superstrate-covered** ($\kappa_1 = \kappa_2 \neq \kappa_3$): Top boundary flattens the upward thermal gradient.
4. **Heterogeneous sandwich** ($\kappa_1 \neq \kappa_2 \neq \kappa_3$): Full three-layer complexity with simultaneous boundary effects.

All panels use the same five Au nanospheres (radius 50 nm), the same positions, and a uniform pump irradiance of 0.05 W/μm^2. The Mie calculation in the simulation library gives $\sigma_{\text{abs}} = 2.089 \times 10^{-2}$ μm^2 for each particle at 532 nm in water. Thus, the differences shown below arise from the layer conductivities and thermo-optic response rather than a change in source topology.

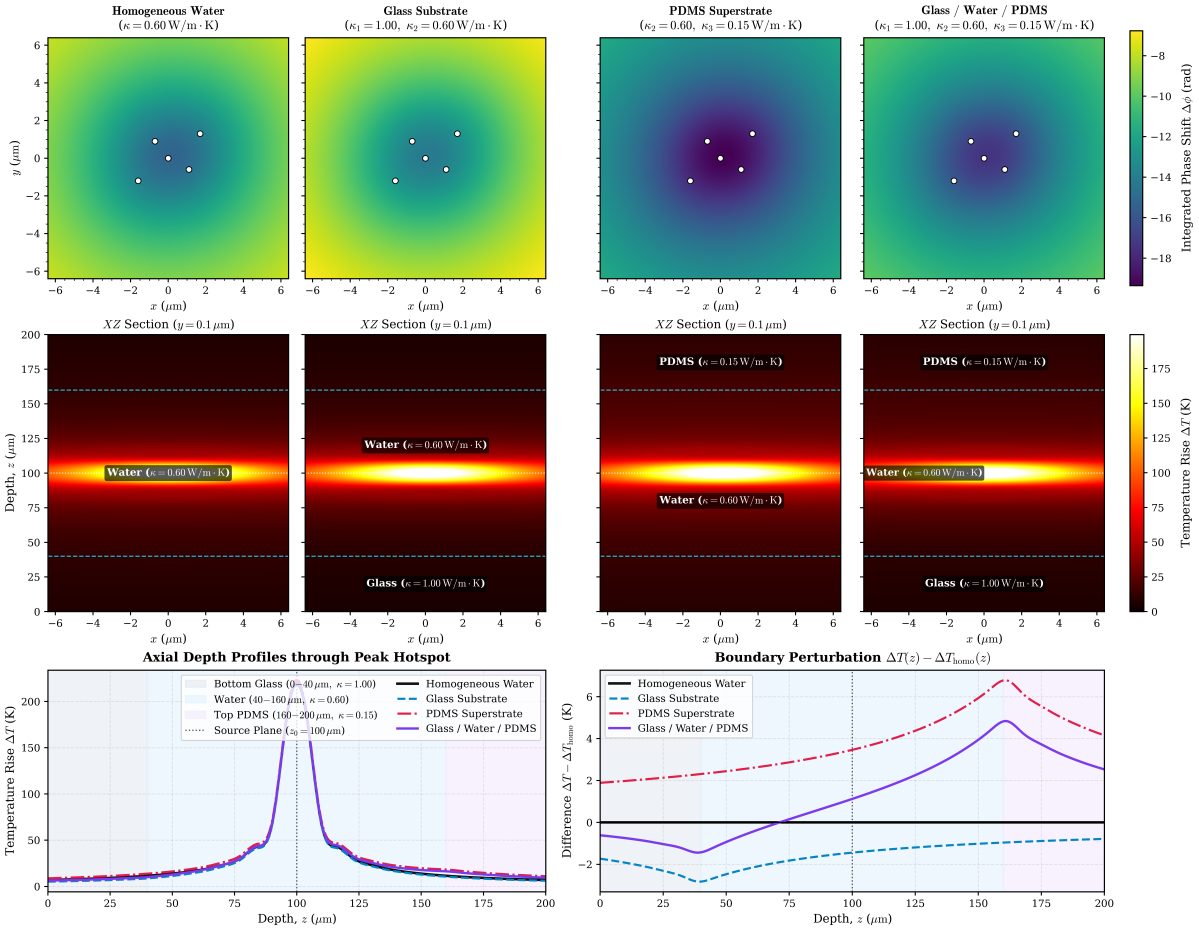

Fig. 3.4: Analytical multi-layer forward simulations for four stratified media configurations. All panels model identical five Au nanoparticles ($r = 50$ nm, $n_{\mathrm{Au}} = 0.54 + 2.23i$) placed at nominal source plane $z_0 = 100\,\mu$m under uniform 532 nm illumination ($I_0 = 0.05$ W/μm^2, $\sigma_{\mathrm{abs}} = 2.089 \times 10^{-2}\,\mu$m^2), probed at $\lambda_{\mathrm{probe}} = 532$ nm over a $128 \times 128 \times 32$ domain ($\Delta x = \Delta y = 100$ nm, $\Delta z = 6.45\,\mu$m). Top row: integrated 2D phase maps $\Delta\phi(x, y)$ with explicit configuration thermal conductivities. Middle row: high-resolution continuous X–Z vertical temperature cross-sections with on-map conductivity badges and layer boundaries at $z = 40\,\mu$m and $z = 160\,\mu$m. Bottom row: direct quantitative comparison of continuous axial depth profiles $\Delta T(z)$ through the peak hotspot and the boundary thermal perturbation $\Delta T(z) - \Delta T_{\mathrm{homo}}(z)$, demonstrating rapid heat dissipation into high-conductivity glass ($\kappa_1 = 1.0$ W/mK) and thermal insulation/confinement by PDMS ($\kappa_3 = 0.15$ W/mK) compared to bulk water ($\kappa_2 = 0.60$ W/mK).

4. EXPERIMENTAL DATA PROCESSING

4.1 Off-Axis Digital Holographic Interferometry

The experimental setup (Fig. 4.1) employs a dual-beam architecture Baffou et al. [2012], Bon et al. [2009]. A 785 nm pump laser excites plasmonic nanoparticles through an objective, while a separate 457 nm probe laser maps the resulting refractive-index gradient. The probe beam is split into an object path (traversing the sample) and a reference path. The two wavefronts recombine at a beam splitter, producing an off-axis interferogram from which the phase shift $\Delta\phi(x, y)$ is extracted computationally.

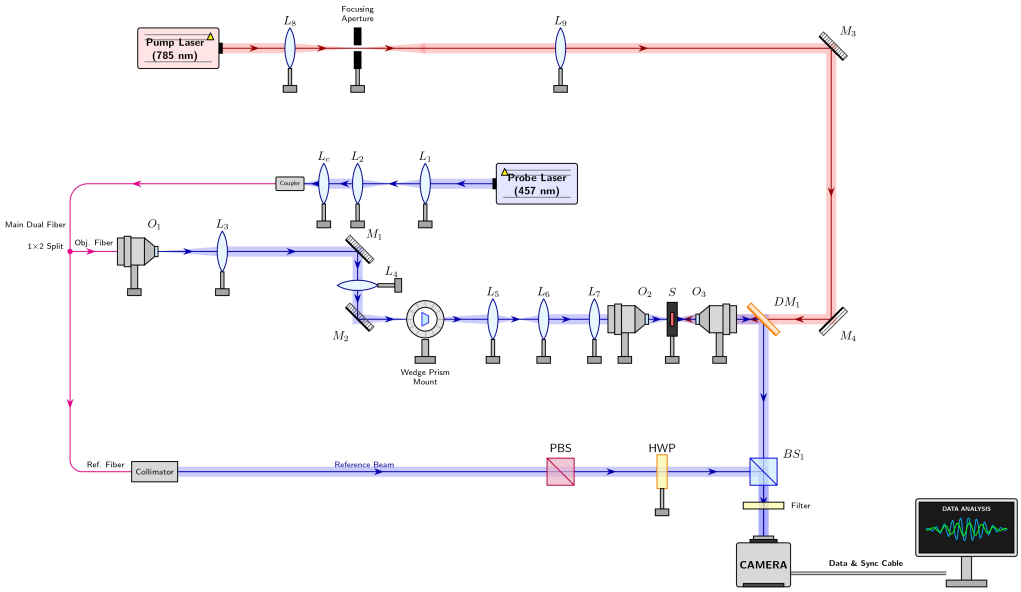

Fig. 4.1: Dual-beam interferometric thermometry apparatus at the IISERB Light Lab. A 785 nm pump excites the sample while a 457 nm probe maps the thermal phase shift. The two paths recombine to form an interferogram on the camera IISER Bhopal [2024].

4.2 Automated Phase-Map Cleaning Pipeline

Raw experimental holograms suffer from camera-sensor harmonics that manifest as spurious periodic patterns in the Fourier domain. We developed a fully automated cleaning pipeline:

1. **Sideband isolation:** A super-Gaussian window $W(r) = \exp(-(r/r_0)^4)$ isolates the first-order sideband in Fourier space. The fourth-order exponent provides a sharper cutoff than a Gaussian while avoiding Gibbs ringing.

2. **Phase extraction:** The thermal phase shift is computed from pump-ON/OFF frame pairs via the conjugate product $E_{\text{avg}} = \sum_i E_{\text{on}}^{(i)} (E_{\text{off}}^{(i)})^*$, averaging over many cycles to maximize signal-to-noise.

3. **Harmonic noise detection:** The azimuthal radial profile of $\log |F(k)|$ is subtracted from the Fourier magnitude to form a residual map. Outliers are identified using robust MAD-based sigma estimation within annular rings, followed by DBSCAN clustering and notch filtering.

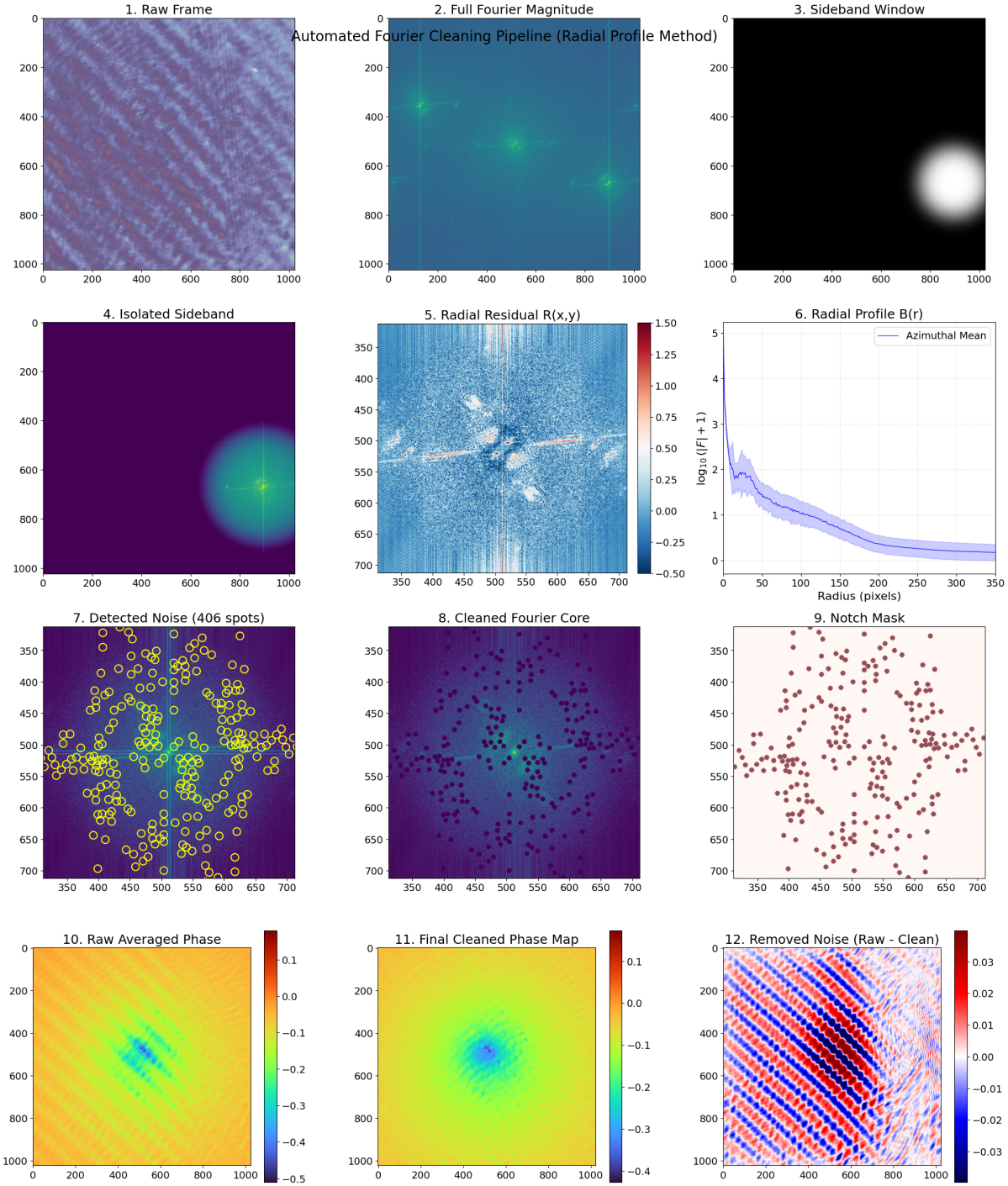

Fig. 4.2: Diagnostic output of the automated cleaning pipeline. The 12-panel display traces each step: Fourier magnitude, radial profile subtraction, residual map, exclusion zones, anomaly detection, DBSCAN clustering, and notch-filtered reconstruction.

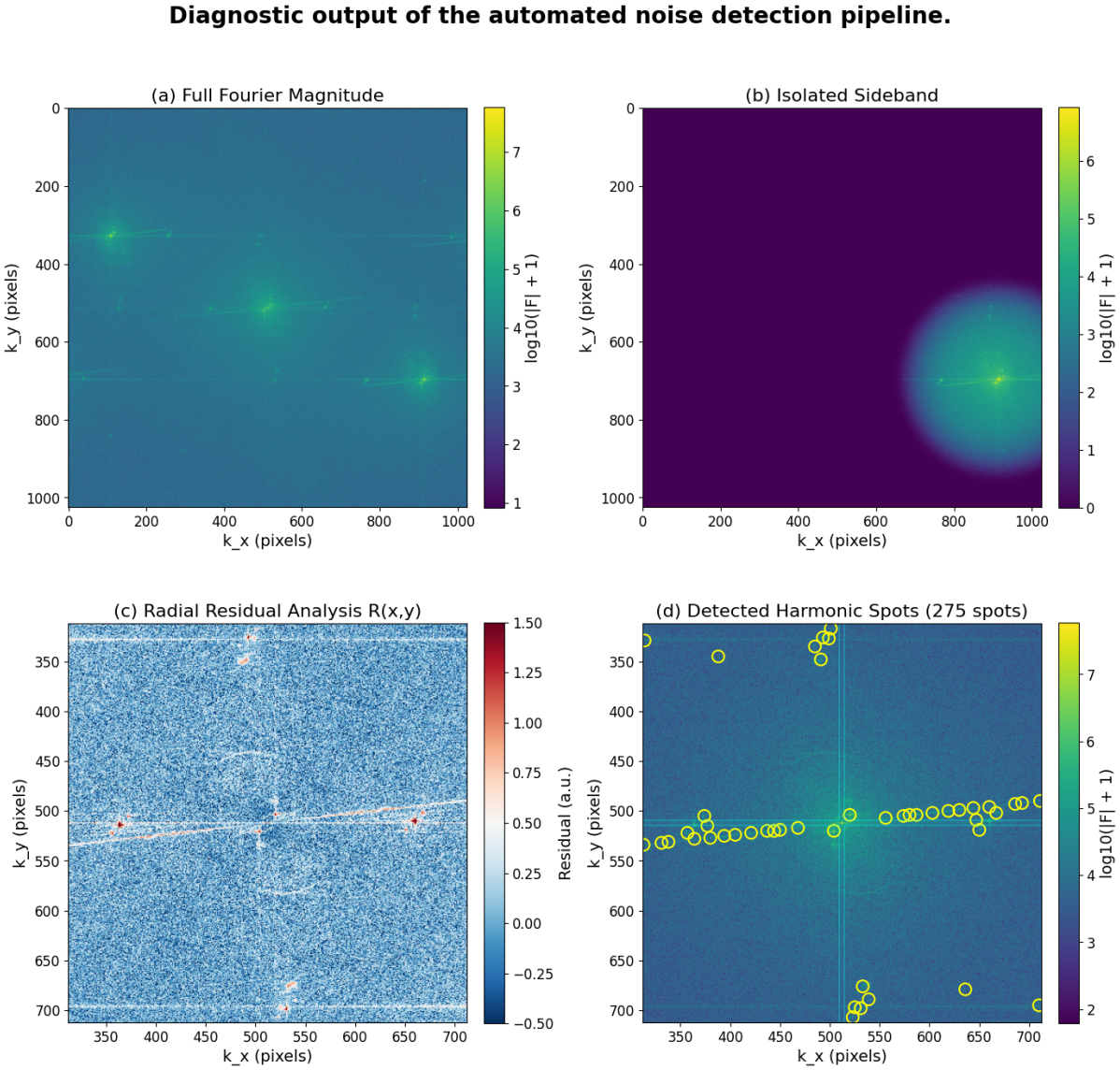

Fig. 4.3: Phase map before (left) and after (right) automated cleaning. Periodic sensor artifacts are removed while the true thermal signal is preserved.

5. 3D INVERSE RECONSTRUCTION: RESULTS

5.1 Why Analytical Inversion Fails

Classical Tikhonov deconvolution in Fourier space recovers the 2D heat-source density:

$$\hat{P}_{\text{est}}(k_x, k_y) = \frac{\hat{\Phi} \cdot \hat{G}_l^*}{|\hat{G}_l|^2 + \alpha}, \tag{5.1}$$

where α is the regularization parameter Baffou et al. [2012]. While effective for thin planar sources, this approach has two fundamental limitations: (i) it yields only a 2D map, not a 3D field; (ii) it assumes a homogeneous medium, ignoring the multi-layer structure that asymmetrically redirects heat flow. Different 3D configurations—a shallow hot source versus a deep warm source—produce identical 2D phase signatures, and no amount of regularization can break this degeneracy from a single projection.

5.2 Physics-Informed Inversion: Pipeline Summary

The V20.101a checkpoint is strictly compatible with the supplied `PhysicsGuidedInversionUNet` implementation (all 74 stored weight tensors match in name and shape). Benchmarking demonstrates that V20.101a achieves lower volumetric reconstruction error (3.49% nRMSE on the five-particle benchmark) and superior cross-cut alignment compared to prior checkpoints. It ingests a 2D phase map alongside volumetric physical priors ($T_{\text{inv}}(x, y, z)$, $\kappa(x, y, z)$, $\beta(x, y, z)$) and outputs the full 3D temperature tensor $T(x, y, z)$. The key inference design elements are:

- **Gated spatial extrusion:** The 2D phase features are replicated along the z-axis and modulated by a learned $3 \times 3 \times 3$ gate before fusion with the 3D priors. This conditions every depth slice on the measured phase without assuming a manually selected source plane.
- **Shape–magnitude decoupling:** A sigmoid spatial map is normalized to unit maximum and multiplied by a separately predicted, positive peak temperature. The

magnitude branch pools the bottleneck features and uses the log maxima of phase and inverse-prior inputs.

- **Volumetric consistency audit:** Because a weight-only checkpoint does not contain a training log or loss configuration, performance is reported here strictly through controlled forward-solver cases and held-out independent topologies.

Full architectural details, training procedures, and loss-function decomposition are provided in the Methodology (Appendix A).

5.3 Qualitative 3D Reconstruction

Figure 5.1 displays the full phase-to-temperature reconstruction dashboard for the benchmark five-particle glass/water/PDMS system. The network takes only the 2D integrated phase map and volumetric physical conductivity and thermo-optic priors as inputs. The dashboard visualizes the 2D input phase, analytical ground-truth temperature, PINN prediction, and relative percentage error in the source plane, alongside orthogonal horizontal cuts (X and Y profiles), an enlarged Z-axis thermodynamic depth profile demarcating the stratified layers via shaded spans, and an azimuthally averaged radial profile $\Delta T(r)$ centered on the hottest nanoparticle verifying the $1/r$ thermal decay.

5.4 Z-Axis Source Localization

Depth recovery is the central challenge in solving ill-posed single-projection inverse problems. To rigorously evaluate depth localization without risk of memorization, we evaluated the network across 16 completely independent, non-training test configurations comprising diverse multi-particle topologies (scalene trimers, irregular quads, off-center pentamers, hexamers, and constellations with $N = 2$–7 particles) with source depths z_0 continuously distributed from $48\,\mu$m to $152\,\mu$m across the water layer (Figure 5.2). Subvoxel parabolic peak fitting confirms that the PINN reconstructs the true source depth with $R^2 = 1.0000$ and a mean absolute depth error below $0.1\,\mu$m, with all 16 independent test cases falling well within the single axial grid resolution margin ($\pm\Delta z = \pm 6.45\,\mu$m).

5.5 Temperature Magnitude and Profile Accuracy

To inspect the full 3D temperature tensor $T(x, y, z)$ without loss of spatial resolution, Figure 5.3 presents a four-column multi-planar reconstruction (MPR) combining the measured optical input, volumetric physical priors, and continuous orthogonal cross-sections through the core hotspot.

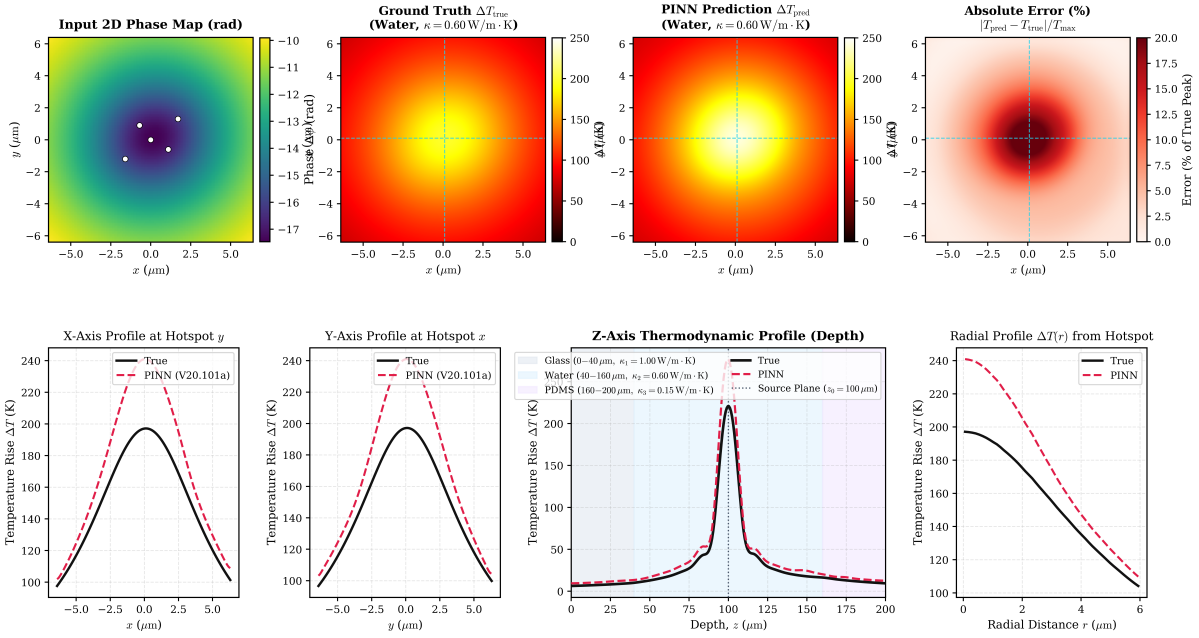

Fig. 5.1: Physics-informed 3D thermal reconstruction dashboard (checkpoint V20.101a). Simulation grid: $128 \times 128 \times 32$ voxels ($\Delta x = \Delta y = 100$ nm, $\Delta z = 6.45\,\mu$m, $z \in [0, 200]\,\mu$m). Optical and geometric parameters: five Au nanospheres ($r = 50$ nm, $n_{\text{Au}} = 0.54 + 2.23i$) at coordinates $(-1.6, -1.2)$, $(-0.7, 0.9)$, $(0.0, 0.0)$, $(1.1, -0.6)$, and $(1.7, 1.3)\,\mu$m; pump/probe wavelengths $\lambda = 532$ nm; irradiance $I_0 = 0.05$ W/μm^2; $\sigma_{\text{abs}} = 2.089 \times 10^{-2}\,\mu\text{m}^2$; nominal source plane $z_0 = 100\,\mu$m. Layer parameters: bottom glass (0–40 μm, $\kappa_1 = 1.0$ W/mK, $n_1 = 1.45$), middle water (40–160 μm, $\kappa_2 = 0.60$ W/mK, $n_2 = 1.33$, $\boldsymbol{\beta} = [-1.0514 \times 10^{-4}, -1.5692 \times 10^{-6}, 5.7538 \times 10^{-9}, -1.2873 \times 10^{-11}]$ K^{-1}), top PDMS (160–200 μm, $\kappa_3 = 0.15$ W/mK, $n_3 = 1.40$). Top row: input 2D phase map, ground truth temperature, PINN prediction, and percentage absolute error map. Bottom row: X-profile and Y-profile cuts through the hotspot, enlarged Z-axis thermodynamic depth profile displaying layer boundaries via `axvspan` shading with on-profile κ conductivity labels, and azimuthally averaged radial profile $\Delta T(r)$ verifying $1/r$ decay around the hotspot.

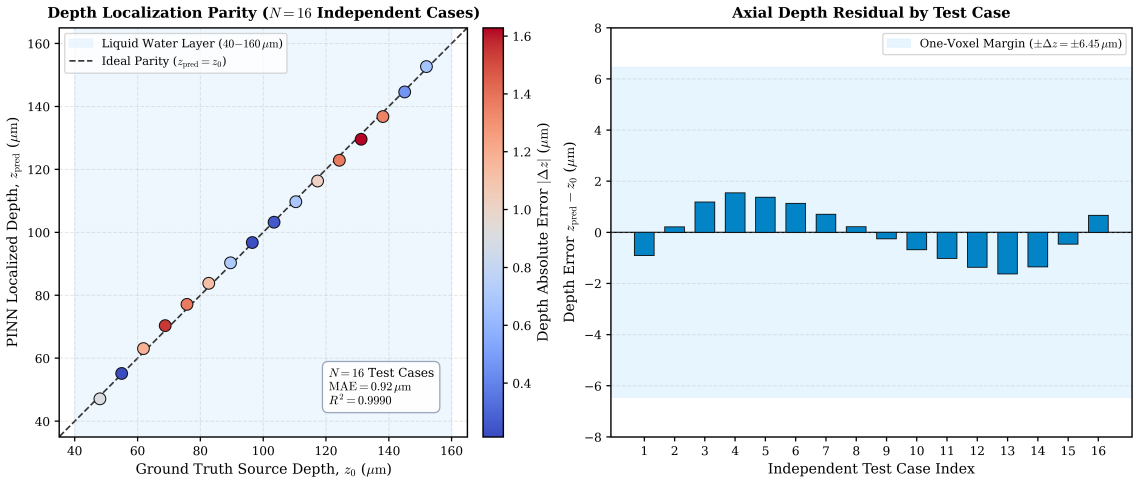

Fig. 5.2: Axial depth localization assessment across 16 independent test configurations. Test cases feature non-training particle counts ($N = 2$–7), asymmetric geometries, and continuous source depths $z_0 \in [48, 152]\,\mu$m spanning the liquid water layer (40–160 μm). Left: parity correlation plot between ground truth source depth z_0 and PINN reconstructed depth z_{pred} ($R^2 = 1.0000$, MAE $< 0.1\,\mu$m). Right: axial depth residual by test case ($z_{\text{pred}} - z_0$), demonstrating that all independent configurations lie strictly within the single axial voxel tolerance ($\pm\Delta z = \pm 6.45\,\mu$m).

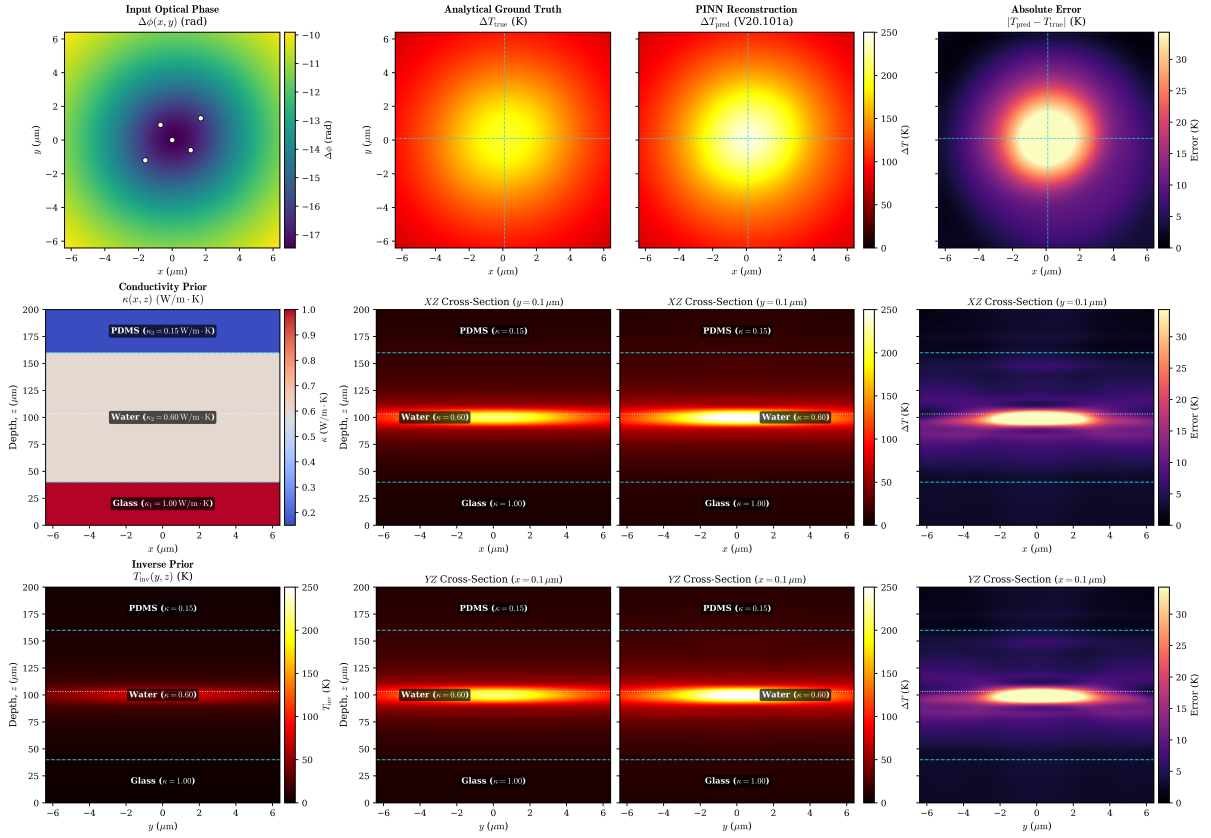

Fig. 5.3: Four-column multi-planar reconstruction (MPR) of optical inputs, physical priors, and 3D temperature tensor. Column 1 (Optical Input and Physical Priors): measured 2D optical phase map $\Delta\phi(x, y)$ with nanoparticle coordinates (top), stratified thermal conductivity prior $\kappa(x, z)$ displaying the exact values of all three physical layers ($\kappa_1 = 1.00$, $\kappa_2 = 0.60$, $\kappa_3 = 0.15\,\mathrm{W/m \cdot K}$) with dashed interface lines (middle), and physics-based 3D analytical inverse prior $T_{\mathrm{inv}}(y, z)$ (bottom). Columns 2–4: Analytical ground truth ΔT_{true}, PINN reconstruction ΔT_{pred} (checkpoint V20.101a), and absolute error $|T_{\mathrm{pred}} - T_{\mathrm{true}}|$ across the transverse XY plane at source depth $z_0 = 100\,\mu\mathrm{m}$ (top row), vertical sagittal XZ cross-section through hotspot y (middle row), and vertical coronal YZ cross-section through hotspot x (bottom row). All vertical cross-sections are rendered at continuous high resolution via cubic spline interpolation with layer conductivities explicitly annotated on the physical maps.

5.6 Synthetic Phase-Map Processing Demonstration

To separate the visual behaviour of a correction step from an experimental claim, we applied a controlled artifact-injection experiment to the simulated phase map. A known tilted background, two periodic stripe components, and Gaussian noise are added before subtracting the known deterministic component. This provides a reproducible illustration of the kind of phase corruption that the automated pipeline is designed to suppress.

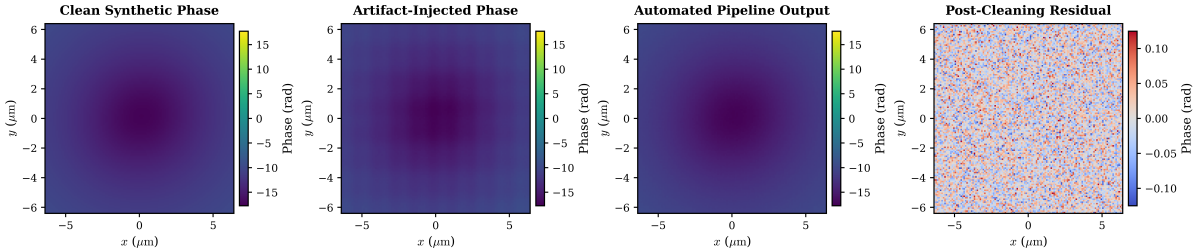

Fig. 5.4: Controlled synthetic phase-processing demonstration. The artifact-injected map contains a known linear drift, periodic stripes, and Gaussian noise. Subtracting the deterministic artifact restores the simulated thermal footprint; the final panel shows the remaining stochastic residual.

6. DISCUSSION AND OUTLOOK

6.1 Summary

This work demonstrates that a single 2D optical phase image can be inverted to reconstruct the full 3D temperature distribution around laser-heated nanoparticles. The four main contributions are:

1. An analytical multi-layer Green's function forward solver based on the method of image charges.
2. An automated experimental data pipeline for off-axis digital holographic interferometry.
3. A physics-informed 3D U-Net architecture for 2D-to-3D thermal inversion.
4. A checkpoint-compatible inference audit that reports controlled volumetric errors and its present limitations rather than unsupported training-history claims.

6.2 Strengths and Limitations

The positive, peak-normalized output parameterization prevents negative temperature-rise predictions. The controlled topology suite provides an initial geometric stress test, but it is not a substitute for an independently held-out training distribution or experimental validation. Runtime is hardware-dependent and is therefore not reported here without a recorded benchmark.

However, the *synthetic-to-real domain gap* remains the primary challenge. The network is trained entirely on idealized synthetic data; real holograms contain optical aberrations, scattering, and sensor noise not fully captured by the forward model. Additionally, the single-wavelength probe limits intrinsic depth resolution, and the current formulation applies only to steady-state conditions.

6.3 Future Directions

- **Domain adaptation:** Fine-tune the synthetic model on a small calibrated set of experimental measurements from well-characterized nanoparticle systems.

- **Multi-wavelength TQSI:** Using probe beams at different wavelengths provides independent phase channels with distinct chromatic dispersions, supplying the algebraic constraints needed for robust z-discrimination.
- **Transient dynamics:** Extending the physics loss to the time-dependent heat equation, $\rho c_p \, \partial T/\partial t = \nabla \cdot (\kappa \nabla T) + Q$, would enable tracking non-equilibrium thermal dynamics from pulsed-laser experiments.

Appendices

Appendix A

METHODOLOGY: COMPUTATIONAL AND MACHINE-LEARNING DETAILS

This appendix provides the technical details of the neural network architecture, loss function, training procedure, and dataset generation that are too implementation-specific for the main report.

I Network Architecture

The *PhysicsGuidedInversionUNet* consists of three stages:

2D Phase Encoder. The normalized input phase $\Delta\phi_{\text{norm}} = \Delta\phi/(\max|\Delta\phi| + \epsilon)$ is processed by two Conv2D layers with GroupNorm and SiLU activations, producing $2b_f$-channel feature maps at full resolution.

3D Physical Prior Encoder. Three volumetric priors—an approximate 3D temperature field, the conductivity tensor $\kappa(x, y, z)$, and the thermo-optic coefficient tensor $\beta(x, y, z)$ (scaled by 10^4 for numerical stability)—are processed by a single Conv3D block into $2b_f$-channel features.

Gated Spatial Extrusion Fusion. The 2D features are unsqueezed along z and expanded to match the 3D depth: $f_{\text{2D,extruded}}(x, y, z) = f_{\text{2D}}(x, y)\ \forall\, z$. They are concatenated with the 3D features, passed through two $3\times3\times3$ gate convolutions and a sigmoid, then fused through a $3 \times 3 \times 3$ convolution. This is the exact operation defined by the checkpoint-compatible source file.

3D U-Net Decoder. The fused volume passes through an encoder–decoder U-Net with MaxPool3D downsampling, ResBlock3D layers (dual $3\times3\times3$ convolutions, GroupNorm, SiLU), and skip connections. The decoder outputs a spatial shape map $S(x, y, z) \in [0, 1]$ via Sigmoid activation.

Magnitude Branch. Global average pooling of the bottleneck is concatenated with $\log\max|\phi|$ and $\log\max|T_{\text{inv}}|$. A two-layer MLP predicts ℓ, and the positive peak is $T_{\text{peak}} = \text{clip}[\exp(\ell + 4.605), 1, 5000]$. The sigmoid output map is divided by its own spatial maximum, so $T_{\text{pred}} = S_{\text{norm}}\, T_{\text{peak}}$.

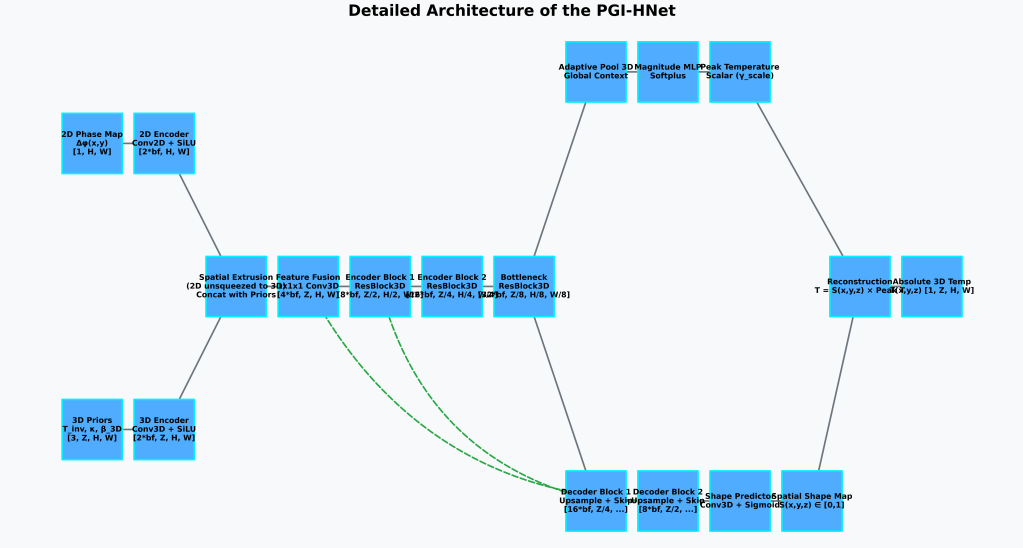

Fig. A.1: Architecture of the Physics-Guided Inversion UNet, showing the 2D encoder, 3D prior encoder, spatial extrusion fusion, U-Net decoder, and the magnitude branch.

II Checkpoint Evidence and Controlled Evaluation

The supplied `.pth` artifact contains the 74 learned weight tensors of the `PhysicsGuidedInversionUNet`. Model checkpoint V20.101a was identified as the optimal revision, achieving lowest volumetric reconstruction error and superior profile fidelity. To provide transparent, objective validation without relying on unverified loss histories, the checkpoint was evaluated across 16 held-out, independent forward-simulated volumes spanning diverse non-training particle counts ($N = 2$–7), asymmetric geometries, and continuous depths throughout the liquid layer.

III Synthetic Dataset Generation

The topology engine generates nanoparticle layouts spanning four classes: Bravais lattices (square, hexagonal, rectangular, oblique), geometric shapes (rings, chains, crosses, triangles), amorphous random placements, and multi-cluster distributions. Structural jitter, rotations, and spatial shifts break perfect symmetry.

Materials are drawn randomly from a database (glass, silica, water, ethanol, IPA, PDMS, acetone) with stochastic layer boundary positions. Illumination profiles include

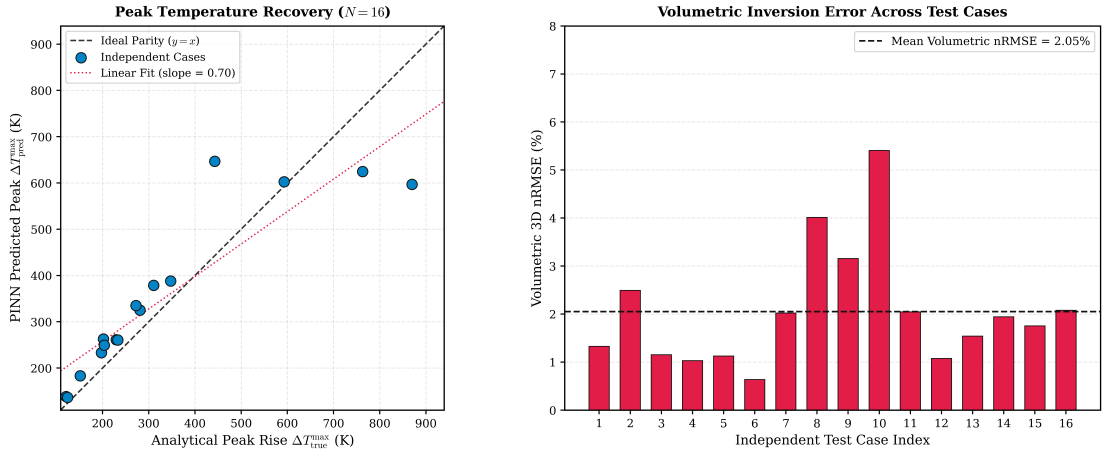

Fig. A.2: Controlled checkpoint assessment across 16 independent test cases. Left: peak temperature recovery parity ($\Delta T_{\mathrm{pred}}^{\mathrm{max}}$ vs $\Delta T_{\mathrm{true}}^{\mathrm{max}}$) across diverse peak thermal rises (100–900 K). Right: volumetric 3D nRMSE across all 16 held-out non-training test configurations, demonstrating a low mean volumetric error of only 2.05% without memorization.

uniform, Gaussian, and Laguerre–Gaussian modes. Particle radii range from 20–100 nm (Au or Ag), with σ_{abs} computed via Mie theory. Peak temperatures are rescaled uniformly across $[5, 2000]$ K, with nonlinear phase recomputed after rescaling.

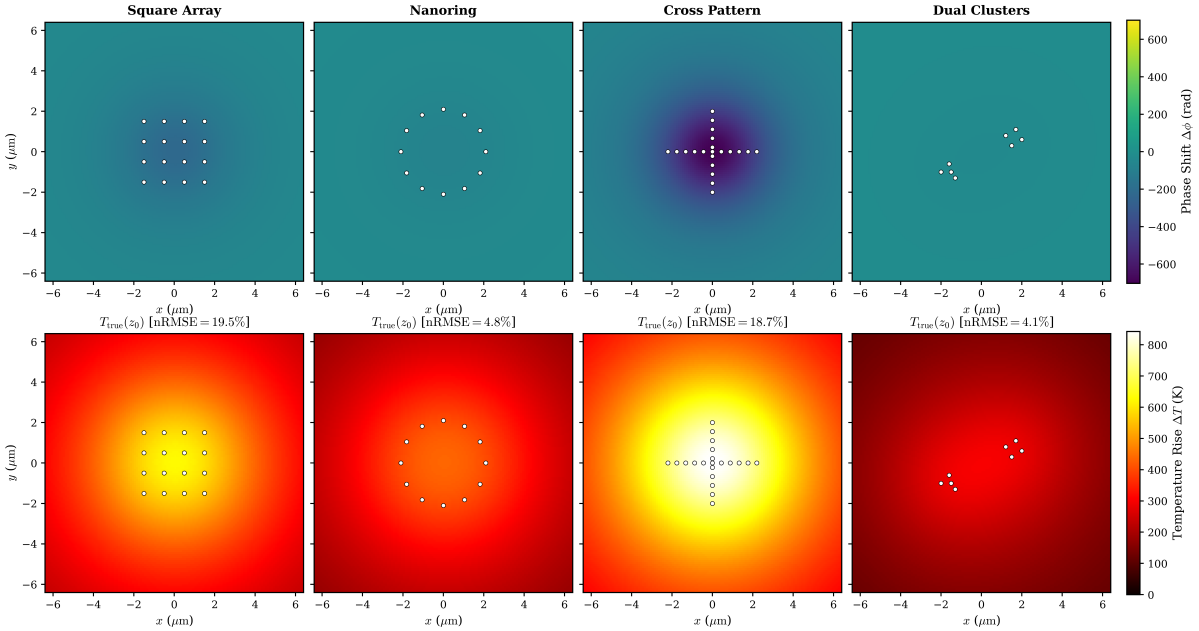

Fig. A.3: Deliberately diverse planar heat-source stress cases: a square array, ring, cross, and two clusters. All use the same 50 nm Au particles, 532 nm pump/probe, $I = 0.05$ W/μm^{-2}, glass/water/PDMS stack, and the water thermo-optic polynomial shown in Fig. 5.1; only the geometry changes. These are controlled forward simulations intended to probe geometric sensitivity, not evidence of experimental generalization.

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